

EXPERIMENT-6

Q1. Write a C program to find the area, perimeter, and diagonal of a rectangle when its length and breadth are given.

(A) Problem Statement:

Write a C program to calculate the area, perimeter, and diagonal of a rectangle using its length and breadth.

Input:

- Length (l)
- Breadth (b)

Output;

- Area
- Perimeter
- Diagonal

Formulae:

- Area = Length $\times$ Breadth
- Perimeter = $2 \times (\text{Length} + \text{Breadth})$
- Diagonal = $\sqrt{(\text{Length}^2 + \text{Breadth}^2)}$

(B) Standard Case Table:

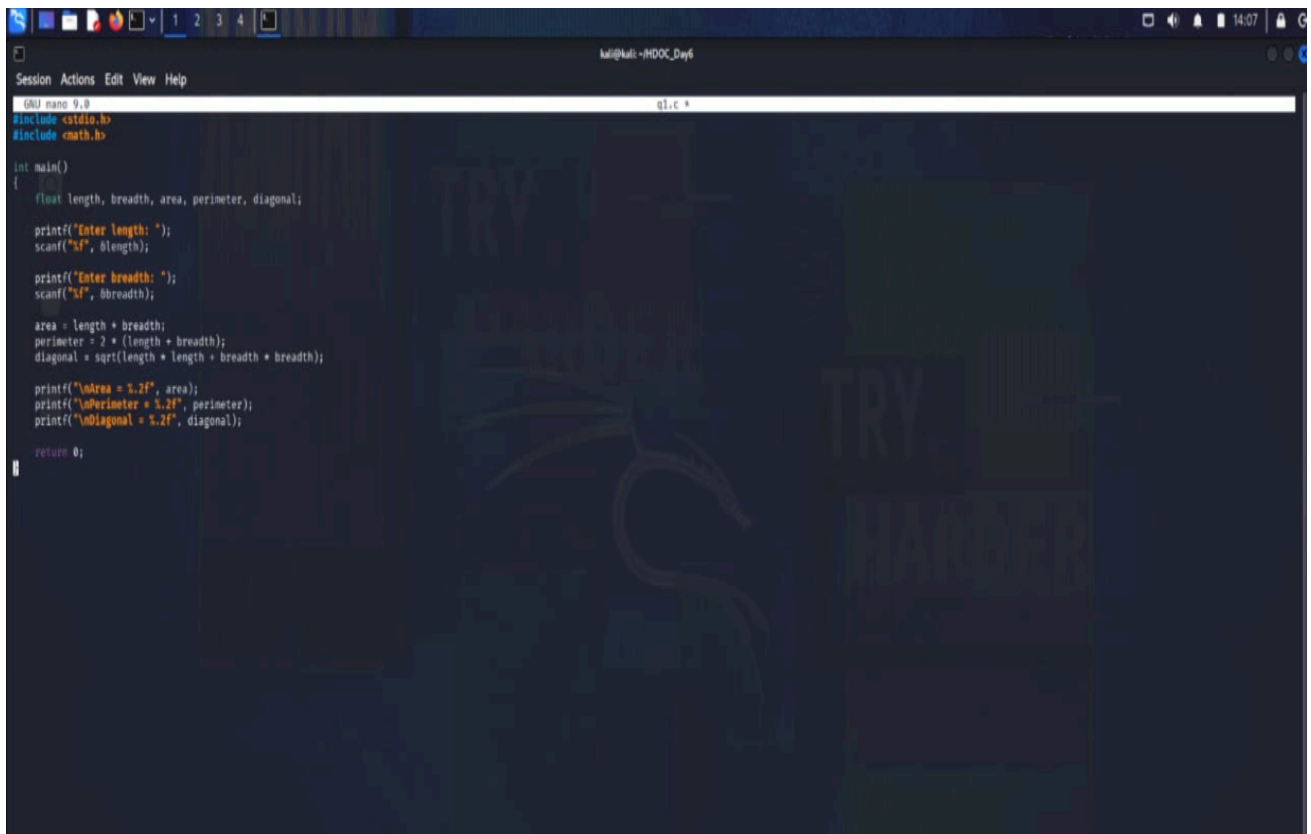
Test Case	Input (Length, Breadth)	Expected Output	Actual Output	Result
1	10, 5	Area = 50, Perimeter = 30, Diagonal = 11.18	Same	Pass

2	8, 6	Area = 48, Perimeter = 28, Diagonal = 10.00	Same	Pass
3	15, 12	Area = 180, Perimeter = 54, Diagonal = 19.21	Same	Pass

(C)Algorithm:

1. Start.
2. Declare variables for length, breadth, area, perimeter, and diagonal.
3. Read the values of length and breadth.
4. Calculate area, perimeter, diagonal.
5. Display the area, perimeter, and diagonal.
6. Stop.

(D)Program:



```

kali@kali:~/HDOOC_Day6
Session Actions Edit View Help
GNU nano 9.8 q1.c
#include <stdio.h>
#include <math.h>

int main()
{
    float length, breadth, area, perimeter, diagonal;

    printf("Enter length: ");
    scanf("%f", &length);

    printf("Enter breadth: ");
    scanf("%f", &breadth);

    area = length * breadth;
    perimeter = 2 * (length + breadth);
    diagonal = sqrt(length * length + breadth * breadth);

    printf("\nArea = %.2f", area);
    printf("\nPerimeter = %.2f", perimeter);
    printf("\nDiagonal = %.2f", diagonal);

    return 0;
}

```

(E)Compilation and Execution Evaluation:

Compilation Status: Successful

Execution Results:

Input:

Length = 10

Breadth = 5

Output:

Area = 50.00

Perimeter = 30.00

Diagonal = 11.18

Hence, The program works right without any problem.

Q2. Write a C program to find one unequal adjacent side, area, and perimeter of a kite when its perimeter, other unequal adjacent side, and two diagonals are given.

(A)Problem Statement:

Write a C program to calculate the missing unequal adjacent side, area, and perimeter of a kite.

Input:

- Perimeter (P)

- One unequal adjacent side (b)
- Diagonal 1 (d_1)
- Diagonal 2 (d_2)

Output:

- Other unequal adjacent side (a)
- Area
- Perimeter

Formulae:

$$\text{Perimeter} = 2(a + b)$$

Therefore,

$$a = (\text{Perimeter} \div 2) - b$$

$$\text{Area} = (d_1 \times d_2) \div 2$$

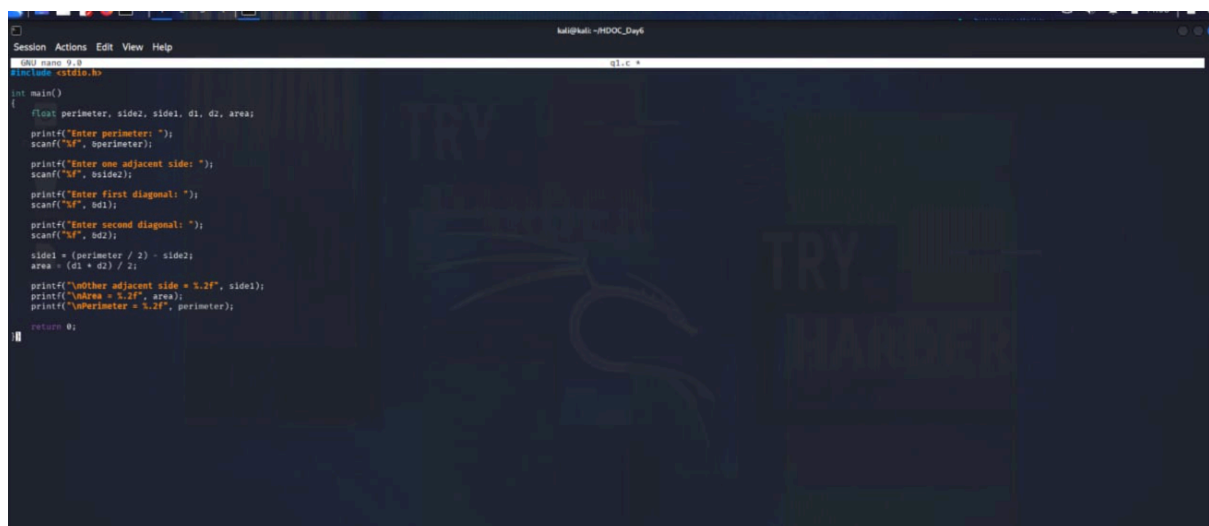
(B)Standard Test Case Table:

Test Case	Input (P, b, d_1 , d_2)	Expected Output	Actual Output	Result
1	40, 8, 10, 12	Side = 12, Area = 60, Perimeter = 40	Same	Pass
2	36, 7, 8, 14	Side = 11, Area = 56, Perimeter = 36	Same	Pass
3	50, 10, 16, 12	Side = 15, Area = 96, Perimeter = 50	Same	Pass

(C) Algorithm:

1. Start.
2. Declare variables for perimeter, one adjacent side, missing adjacent side, diagonals, and area.
3. Read the perimeter, one adjacent side, and the two diagonals.
4. Calculate the missing adjacent side by subtracting the known side from half of the perimeter.
5. Calculate the area using half the product of the diagonals.
6. Display the missing adjacent side, area, and perimeter.
7. Stop.

(D) Program:

A screenshot of a code editor window showing a C program. The editor has a dark theme and a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The file name is 'q1.c'. The code is as follows:

```
int main()
{
    float perimeter, side2, side1, d1, d2, area;
    printf("Enter perimeter: ");
    scanf("%f", &perimeter);
    printf("Enter one adjacent side: ");
    scanf("%f", &side2);
    printf("Enter first diagonal: ");
    scanf("%f", &d1);
    printf("Enter second diagonal: ");
    scanf("%f", &d2);
    side1 = (perimeter / 2) - side2;
    area = (d1 * d2) / 2;
    printf("\nOther adjacent side = %.2f", side1);
    printf("\nArea = %.2f", area);
    printf("\nPerimeter = %.2f", perimeter);
    return 0;
}
```

(E) Compilation and Execution Evaluation:

Compilation Status: Successful

Execution Results:

Input:

Perimeter = 40

One adjacent side = 8

Diagonal 1 = 10

Diagonal 2 = 12

Output:

Other adjacent side = 12.00

Area = 60.00

Perimeter = 40.00

Hence, The program works right without any problem.

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